

SW Engineering CSC 648-848 Fall 2025

Gator Learn

Team #5

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Github Lead	Hill Kalathiya
Frontend Lead	Enrique Joachim Liganor
Backend Lead	Jonah Hammond
Support	Samantha Chombo-Rodriguez

Milestone 1

Date submitted	Date revised
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1 Executive Summary

Ever had an experience where you thought "Oh no, I have an exam in a week, and I don't understand the material!" but don't know where to find support? Gator Learn has your back!! Gator Learn is a student-led academic support platform that's designed to empower San Francisco State University students through accessible, affordable and personalized tutoring exclusively for SFSU students, by SFSU students. Our platform fosters a collaborative learning environment built on shared academic experiences and community trust. We offer one-on-one and group tutoring across a wide range of SFSU courses including STEM, Humanities, Business, Writing, and general education requirements! Our mission is to enhance student success, confidence and retention by making academic support convenient and relatable. Sessions are available in-person and online with flexible scheduling to fit the demands of university life. Beyond coursework help, tutors provide study strategies, exam preparation, and mentorship to help students thrive long term!

Gator Learn! Helping Gators support Gators!

2 Personae

Persona 1: Sarah Patel - The Student

About Sarah:

- Age: 20
- Major: Computer Science
- Year: Junior

Attitude:

Curious and motivated, but gets easily overwhelmed by heavy coursework and time constraints. Prefers quick and reliable help.



Skills:

Basic to intermediate tech-savvy, uses mobile and web apps regularly,

Limitations:

Struggles with midterms and assignments, and on a tight budget.

Goals:

- Find verified tutors from her own university
- Schedule sessions that easily fit her timetable
- Get an affordable tutor to improve the grades..

Pain Points:

- Difficult to find reliable peer tutors quickly
- Limited budget and availability conflict

Persona 2: James Lee - The Tutor

About James:

- Age : 21
- Major: Computer Science
- Year: Senior

Attitude:

Eager to help peers while gaining teaching experience and earn some money.

Skills:

Expertise in specific subjects, and uses digital tools regularly.

Limitations:

Busy schedule balancing coursework, tutoring, and researching

Pain Points:

- Inconsistent tutoring request hours.
- Lengthy verification process for a tutor

Goals:

- Earn extra income through tutoring.



- Build teaching experience for graduate school.
- Connect with motivated students on campus.

Persona 3: Emily Roberts - The Admin

About Emily:

- Role: Academic Success Coordinator
- Age: 38

Attitude:

Responsible for maintaining the quality and smooth operation of the tutoring platform

Skills:

Manages user accounts, approves tutor listing applications, and oversees system functionality

Limitations:

Managing approval workflows and user issues promptly.

Pain Points:

- Numerous pending tutor listing applications.
- Limited tools to assess tutor performance efficiently.

Goals:

- Approve and monitor qualified tutors.
- Maintain platform integrity and trust.



Persona 4: Alex Gomez - The Unregistered User

About Alex:

- Age: 18
- Year: Freshmen

Attitude:

Curious, skeptical, and exploring options before committing.

Skills:

Limited to basic web navigation, browsing, and research.

Limitations:

Cannot book the tutors without logging in.

Pain points:

- Unclear service details without registration.
- Concerned about sharing personal university credentials.

Goals:

- Learn what tutoring services are available.
- Understand pricing and reliability before signing up.



3 High-Level Use Cases

Use Case 1: Student registers, looks for a tutor, and messages the tutor.

Sarah Patel hears about the new SF State tutoring app from a flyer on campus. Sarah is desperate and stressed as she hasn't been studying consistently. She browses around the site in hopes of finding a tutor in her weakest subject. She finds multiple tutor listings on the subject, but she doesn't know which one is cheapest, reliable, or fits her schedule, so she filters by price from lowest to highest price and is available to meet on Tuesdays after 2 pm. She finds a tutor who is reasonably priced and can meet at a later time on Tuesdays at the library. She messages the tutor in the app her contact information that she wishes to share, then the site asks her to register before sending. She registers via student email to confirm she is a student at SFSU. She finally sends the message, and the rest is history. Tutor and her continue conversation off-site. Sarah finds at least one tutor matching price/time/location and gets connected.

Use Case 2: Student wants to post a tutor listing

James Lee hears from other students that they are getting paid to tutor on the SF State tutor app. James does well in the majority of his courses and is looking for a way to make money. He browses the site through courses that don't have as many tutors as he is comfortable with. He fills out the tutor listing form and can attach his CV to his tutor listing. Before submission, he is prompted to sign up as a student to be granted access to the tutor listing sign-up. He enters his student credentials to create a student account, then can post a tutor listing. He clicks the button to post a listing for every course he thinks he is qualified to tutor for. James has to wait until the admin accepts his listings.

Use Case 3: Student uses the dashboard

James Lee waited a day to see if his tutor listings were accepted. In order to see if he is accepted, he signs in as student. He is informed by the system that he is accepted. He signs in and sees his tutor dashboard for any messages. He gets a lot of messages in one go as it's midterms time. He wants to respond to all of them, but he thinks he can only take up to three students. He responds to three of the students' messages off the app to not overwhelm himself. He proceeds to tutor and gets paid by the students off-site.

Use Case 4: Admin responsibilities for a safe app

Emily Roberts was hired to monitor the app, ensuring students and tutors were behaving properly. She gets to check pending posts put up by tutors to ensure nobody posts inappropriate content. She can deny or accept any listing posted before they are live on-site. Furthermore, she can check the quality of the tutor via the listing and post a CV on the listing. Emily receives reports from the public support email on-site in case of inappropriate behaviors from user accounts. She can remove accounts and listings of users who have been reported. Emily ensures pending posts get processed, and foul acts are restricted from the site.

4 Data Glossary

User Types

1. **Unregistered User** - A visitor who can view general information but cannot interact with or book any tutoring services. They can browse and sort through the tutoring listings and view the home page
2. **Registered User** - A verified university student who has completed registration. They are the main consumers of tutoring services. They inherit the permissions of unregistered users, in addition to being able to respond to tutor listings and create and edit their own. They also have access to a dashboard of all messages and listings
3. **Admin** - Authorized university staff responsible for platform integrity, tutor verification, and issue resolution. They manage the content of the website. They inherit the permissions of the registered users. They approve or reject tutor listing applications and moderate users and listings

Key Data Entities

1. UserAccount:

Description: Stores basic information for every individual using the platform , including students, tutors, and admins.

Attributes:

1. Name
2. E-mail
3. Password

Usage: Used for login, access control, and permission management. Acts as the foundational link between all user-related data (profiles, sessions, messages).

2. TutorListing:

Description: Represents a tutor's service offering

Attributes:

1. Student Name
2. Major
3. Subject
4. Price per hour
5. Available time(s)
6. Photo (Optional)
7. Resume (CV)
8. Description
9. Tutoring Video Sample (Optional)
10. Live (True/False)

Usage: Displayed to students during tutor search and booking. It helps students to compare available tutors based on price, subject, and timing before making a reservation.

3. Message:

Description: A one-way communication channel where a student sends a message to a listing owner

Attribute:

1. Student Name
2. Listing Owner Name
3. Tutor Listing
4. Message Content
5. Date and Time
6. Phone Number (Optional)

Usage: Used by the listed student who is getting the tutoring request to get the information about the student, including phone number, and a message..

4. Subject:

Description: Represents an SFSU academic subject that tutors can teach and students can search for.

Attribute:

1. Subject Name
2. Category
3. Description

Usage: Used to categorize tutor listings and organize search filters by subject area. Supports subject-based matching during tutor listing browsing.

5. Course:

Description: Represents an SFSU academic course that tutors can teach and students can search for

Attribute:

1. Course Name
2. Subject
3. Course Number
4. Description

Usage: Used to categorize tutor listings and organize search filters by course. Supports course-based matching during tutor listing browsing.

5 Functional Requirements

Unregistered Users

1. Unregistered users shall be able to register on a dedicated registration page with required name and @sfsu.edu email and optional photo, description, and pronouns
2. Unregistered users shall be able to search and sort tutor listings by name, time, price, SFSU subject, and SFSU course number

Registered Users

3. Registered users shall inherit all of the above requirements and permissions
4. Registered users shall be able to login on a dedicated login page
5. Registered users shall be able to logout
6. Registered users shall be allowed to edit account information after registering
7. Registered users shall be able to reply to tutor listings with in-site messaging system
8. Registered users shall have be able to use dashboard that contains messages received, currently created tutor listings, and pending applications
9. Registered users shall be able to submit tutor listing applications with required title, description, price, and CV and optional photo and video for post approval
10. Registered users shall be able to delete their tutor listings or applications

Administration

11. Administrators shall inherit all of the above requirements and permissions
12. Administrator account registrations shall be assigned by manually website manager
13. Administrators shall be required to review all tutor listings before they go live
14. Administrators shall be able to view website analytics
15. Administrators shall be able to remove any registered user or tutor listing

6 Non-Functional Requirements

1. Application shall be developed, tested and deployed using tools and servers approved by Class CTO and as agreed in M0
2. Application shall be optimized for standard desktop/laptop browsers e.g. must render correctly on the two latest versions of two major browsers
3. All or selected application functions shall render well on mobile devices (no native app to be developed)
4. Posting of tutor information and messaging to tutors shall be limited only to SFSU students
5. Critical data shall be stored in the database on the team's deployment server.
6. No more than 50 concurrent users shall be accessing the application at any time
7. Privacy of users shall be protected
8. The language used shall be English (no localization needed)
9. Application shall be very easy to use and intuitive
10. Application shall follow established architecture patterns
11. Application code and its repository shall be easy to inspect and maintain
12. Google analytics shall be used
13. No e-mail clients shall be allowed. Interested users (clients) can only message to service providers via in-site messaging. One round of messaging (from client to service provider) is enough for this application. No chat functions shall be developed or integrated
14. Pay functionality (e.g. paying for goods and services) shall not be implemented nor simulated in UI.
15. Site security: basic best practices shall be applied (as covered in the class) for main data items
16. Media formats shall be standard as used in the market today
17. Modern SE processes and tools shall be used as specified in the class, including collaborative and continuous SW development and GenAI tools

18. The application UI (WWW and mobile) shall prominently display the following exact text on all pages "SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only" at the top of the WWW page Nav bar. (Important so as to not confuse this with a real application).

7 Competitive Analysis

<u>Feature</u>	<u>Wyzant</u>	<u>Varsity Tutors</u>	<u>Gator Learn</u>
No Additional Fees	-	-	++
Community Led	-	-	++
Search by SFSU Subject	-	-	++
Search by SFSU Course	-	-	++
Tutor Listing Sort	+	-	+
Built-in Messaging System	++	++	+

++ feature is superior; + feature exists; - feature does not exist

Our product, Gator Learn, is a tutor to student connection service exclusively for SFSU students and faculty. Thus, we shall have features which are competitive within our niche market, perhaps most notably a search system allowing discovery of tutors by SFSU subject or course. In addition, we shall implement functionality to sort tutor listings by several categories. As a final note, we shall be competitive in non-technical ways, such as offering no additional fees and being community led.

8 System Architecture and Technologies

Server Host: AWS EC2 (t4g.small: 2 vCPUs & 2 GiB of RAM, Free)

Operating System: Amazon Linux 2023

Database: MySQL v8.4 LTS

WebServer: Nginx 1.28.0

SSL Cert: Let's Encrypt (Cert Bot)

SASS: Dart Sass 1.x

Server-Side Language: Java (Amazon Corretto 21)

Web Framework: Spring Boot 3 (Embedded tomcat)

IDE: IntelliJ IDEA

Web Analytics: Google Analytics 4

Build & Run: Maven

Front-End Language: React JS 18.3

Browsers: Chrome and Firefox

9 GenAI Usage

- Used GenAI for creating personae and the glossary.
Prompts & analysis of responses:
 1. “Give me a persona image for tutor, admin, students, and unregistered guests for a peer-to-peer tutoring website for a university.” (Highly useful).
Would not be able to find better than what Gemini gave me.
 2. “I am working on a Tutoring website for a peer-to-peer tutoring website for a university. Give me personas containing About, Attitude, Skills, Limitation, and Pain point for each” (Moderately useful)
The response from Gemini gave me a great start, and helped me to brainstorm ideas on what could be included. Still there were points I had which were better than ChatGPT for Goals and Pain Points.
 3. “I am working on the data glossary. It should include the Main actors/Users and the Key entities/terms used in the project.” (Minimally useful)
The response from Gemini included many feature related terms (e.g, TutorProfile, Two way messaging between students, listed students, and Notifications) which are not core data entities for this project.

10 Team and Roles

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Support	Samantha Chombo-Rodriguez	schomborodriguez@sfsu.edu

11 Team Lead Checklist

- ✓ So far all team members are fully engaged and attending team sessions when required
- ✓ Team found a time slot to meet outside of the class
- ✓ Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing
- ✓ Team reviewed class slides on requirements and use cases before drafting Milestone 1
- ✓ Team reviewed non-functional requirements from “How to start...” document and developed Milestone 1 consistently
- ✓ Team lead checked Milestone 1 document for quality, completeness, formatting and compliance with instructions before the submission
- ✓ Team lead ensured that all team members read the final M1 and agree/understand it before submission
- ✓ Team shared and discussed experience with GenAI tools among themselves
- ✓ Github organized as discussed in class (e.g. master branch, development branch, folder for milestone documents etc.)